PSO – vzorce

F(x) = p(X < x) · No(μ, σ) · q = 1 − p

Jevy a kombinatorika

$$p(\bar{A}) = 1 - p(A)$$

$$p(A \cup B) = p(A) + p(B) - p(A \cap B)$$

$$p(A \mid B) = \frac{p(A \cap B)}{p(B)}$$

nezávislé: $p(A \cap B) = p(A)\,p(B)$

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}, \qquad V_k(n) = \frac{n!}{(n-k)!}, \qquad V_k'(n) = n^k$$

$$p(A) = \frac{m}{n}, \qquad p(A) = \frac{|A|}{|\Omega|} \text{ (geometrická: délka, plocha)}$$

Úplná pravděpodobnost, Bayes, opakování

$$p(A) = \sum_i p(H_i)\,p(A \mid H_i), \qquad \sum_i p(H_i) = 1$$

$$p(H_k \mid A) = \frac{p(H_k)\,p(A \mid H_k)}{\sum_i p(H_i)\,p(A \mid H_i)}$$

$$p(A \text{ dřív než } B) = \frac{p(A)}{p(A) + p(B)}$$

$$p(\text{začínající vyhraje}) = \frac{a}{1 - (1-a)(1-b)}$$

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}, \qquad \sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$$

Náhodná veličina

$$F(x) = p(X < x), \qquad p(a \le X < b) = F(b) - F(a)$$

$$p(X = x_i) = F(x_i^+) - F(x_i)$$

$$EX = \sum x_i p_i \quad \text{nebo} \quad \int x f(x)\,dx$$

$$EX^2 = \sum x_i^2 p_i \quad \text{nebo} \quad \int x^2 f(x)\,dx$$

$$DX = EX^2 - (EX)^2, \qquad \sigma = \sqrt{DX}$$

$$E(aX + b) = a\,EX + b, \qquad D(aX + b) = a^2 DX$$

$$E(X + Y) = EX + EY; \quad \text{nezávislé: } D(X + Y) = DX + DY$$

$$\int_{-\infty}^{\infty} f(x)\,dx = 1, \qquad f \ge 0, \qquad F(x) = \int_{-\infty}^{x} f(t)\,dt$$

$$p(a < X < b) = \int_a^b f(x)\,dx, \qquad F(\tilde{x}) = 0{,}5$$

Diskrétní rozdělení

	$p(X = k)$	EX	DX
Bi(n, p), k = 0..n	$\binom{n}{k} p^k q^{n-k}$	np	npq
Ge, k = 0, 1, …	$p^k q$	$\frac{p}{q}$	$\frac{p}{q^2}$
Ge, k = 1, 2, …	$q^{k-1} p$	$\frac{1}{p}$	$\frac{q}{p^2}$
H(N, M, n)	$\frac{\binom{M}{k}\binom{N-M}{n-k}}{\binom{N}{n}}$	$n\frac{M}{N}$	$n\frac{M}{N}\frac{N-M}{N}\frac{N-n}{N-1}$
Po(λ), k = 0, 1, …	$\frac{\lambda^k}{k!} e^{-\lambda}$	λ	λ

Bi(n, p) ≈ Po(np) pro velké n, malé p.

Spojitá rozdělení

	$f(x)$	EX	DX
rovnoměrné ⟨a, b⟩	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
exponenciální, x ≥ 0	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Laplace	$\frac{\lambda}{2} e^{-\lambda\lvert x\rvert}$	0	$\frac{2}{\lambda^2}$
No(μ, σ)	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2

Integrály

$$\int x^n\,dx = \frac{x^{n+1}}{n+1}, \qquad \int e^{ax}dx = \frac{e^{ax}}{a}, \qquad \int \frac{dx}{x} = \ln|x|$$

$$\int uv'\,dx = uv - \int u'v\,dx$$

$$\int x \sin x\,dx = \sin x - x\cos x$$

$$\int x \cos x\,dx = \cos x + x \sin x$$

$$\int x^2 \sin x\,dx = -x^2\cos x + 2x\sin x + 2\cos x$$

$$\int x^2 \cos x\,dx = x^2 \sin x + 2x\cos x - 2\sin x$$

$$\int xe^{-\lambda x}dx = -\frac{e^{-\lambda x}}{\lambda^2}(\lambda x + 1)$$

$$\int x^2 e^{-\lambda x}dx = -\frac{e^{-\lambda x}}{\lambda^3}(\lambda^2 x^2 + 2\lambda x + 2)$$

Normální rozdělení

$$u = \frac{x - \mu}{\sigma}, \qquad p(X < x) = \Phi(u)$$

$$\Phi(-u) = 1 - \Phi(u)$$

$$p(a < X < b) = \Phi\left(\tfrac{b-\mu}{\sigma}\right) - \Phi\left(\tfrac{a-\mu}{\sigma}\right)$$

$$x_p = \mu + \sigma u_p, \qquad u_{1-p} = -u_p$$

$$p(|X - \mu| < \sigma u_{(1+\gamma)/2}) = \gamma$$

Limitní věty

$$\mathrm{Bi}(n, p) \approx \mathrm{No}\left(np,\ \sqrt{npq}\right), \qquad npq > 9$$

$$\tfrac{X}{n} \approx \mathrm{No}\left(p,\ \sqrt{pq/n}\right)$$

$$\textstyle\sum_{i=1}^{n} X_i \approx \mathrm{No}\left(n\mu,\ \sigma\sqrt{n}\right), \qquad \bar{X} \approx \mathrm{No}\left(\mu,\ \sigma/\sqrt{n}\right)$$

$$p(|X - EX| \ge \varepsilon) \le \frac{DX}{\varepsilon^2}$$

Φ(u) = p(U < u), U ~ No(0, 1); Φ(−u) = 1 − Φ(u)

u	0,00	0,01	0,02	0,03	0,04	0,05	0,06	0,07	0,08	0,09
0,0	,5000	,5040	,5080	,5120	,5160	,5199	,5239	,5279	,5319	,5359
0,1	,5398	,5438	,5478	,5517	,5557	,5596	,5636	,5675	,5714	,5753
0,2	,5793	,5832	,5871	,5910	,5948	,5987	,6026	,6064	,6103	,6141
0,3	,6179	,6217	,6255	,6293	,6331	,6368	,6406	,6443	,6480	,6517
0,4	,6554	,6591	,6628	,6664	,6700	,6736	,6772	,6808	,6844	,6879
0,5	,6915	,6950	,6985	,7019	,7054	,7088	,7123	,7157	,7190	,7224
0,6	,7257	,7291	,7324	,7357	,7389	,7422	,7454	,7486	,7517	,7549
0,7	,7580	,7611	,7642	,7673	,7704	,7734	,7764	,7794	,7823	,7852
0,8	,7881	,7910	,7939	,7967	,7995	,8023	,8051	,8078	,8106	,8133
0,9	,8159	,8186	,8212	,8238	,8264	,8289	,8315	,8340	,8365	,8389
1,0	,8413	,8438	,8461	,8485	,8508	,8531	,8554	,8577	,8599	,8621
1,1	,8643	,8665	,8686	,8708	,8729	,8749	,8770	,8790	,8810	,8830
1,2	,8849	,8869	,8888	,8907	,8925	,8944	,8962	,8980	,8997	,9015
1,3	,9032	,9049	,9066	,9082	,9099	,9115	,9131	,9147	,9162	,9177
1,4	,9192	,9207	,9222	,9236	,9251	,9265	,9279	,9292	,9306	,9319
1,5	,9332	,9345	,9357	,9370	,9382	,9394	,9406	,9418	,9429	,9441
1,6	,9452	,9463	,9474	,9484	,9495	,9505	,9515	,9525	,9535	,9545
1,7	,9554	,9564	,9573	,9582	,9591	,9599	,9608	,9616	,9625	,9633
1,8	,9641	,9649	,9656	,9664	,9671	,9678	,9686	,9693	,9699	,9706
1,9	,9713	,9719	,9726	,9732	,9738	,9744	,9750	,9756	,9761	,9767
2,0	,9772	,9778	,9783	,9788	,9793	,9798	,9803	,9808	,9812	,9817
2,1	,9821	,9826	,9830	,9834	,9838	,9842	,9846	,9850	,9854	,9857
2,2	,9861	,9864	,9868	,9871	,9875	,9878	,9881	,9884	,9887	,9890
2,3	,9893	,9896	,9898	,9901	,9904	,9906	,9909	,9911	,9913	,9916
2,4	,9918	,9920	,9922	,9925	,9927	,9929	,9931	,9932	,9934	,9936
2,5	,9938	,9940	,9941	,9943	,9945	,9946	,9948	,9949	,9951	,9952
2,6	,9953	,9955	,9956	,9957	,9959	,9960	,9961	,9962	,9963	,9964
2,7	,9965	,9966	,9967	,9968	,9969	,9970	,9971	,9972	,9973	,9974
2,8	,9974	,9975	,9976	,9977	,9977	,9978	,9979	,9979	,9980	,9981
2,9	,9981	,9982	,9982	,9983	,9984	,9984	,9985	,9985	,9986	,9986
3,0	,9987	,9987	,9987	,9988	,9988	,9989	,9989	,9989	,9990	,9990

Hodnoty bez „0“: ,8413 = 0,8413. Φ(3,5) = 0,99977.

Náhodný vektor

$$\iint f(x,y)\,dx\,dy = 1, \qquad p((X,Y)\in G) = \iint_G f(x,y)\,dx\,dy$$

$$f_X(x) = \int f(x,y)\,dy, \qquad f_Y(y) = \int f(x,y)\,dx$$

$$\text{nezávislé} \iff f(x,y) = f_X(x)\,f_Y(y)$$

$$\operatorname{cov}(X,Y) = E(XY) - EX\,EY, \qquad \rho = \frac{\operatorname{cov}(X,Y)}{\sigma_X \sigma_Y}$$

Výběrové charakteristiky

$$\bar{x} = \frac{1}{n}\sum x_i$$

$$s^2 = \frac{1}{n-1}\sum (x_i - \bar{x})^2 = \frac{\sum x_i^2 - (\sum x_i)^2/n}{n-1}$$

$$\text{četnosti: } \bar{x} = \frac{\sum x_j n_j}{n}, \qquad \sigma^2 = \frac{\sum x_j^2 n_j}{n} - \bar{x}^2$$

Intervaly spolehlivosti (1 − α)

$$\mu \in \bar{x} \pm t_{1-\alpha/2}(n-1)\,\frac{s}{\sqrt{n}}$$

$$\mu \in \bar{x} \pm u_{1-\alpha/2}\,\frac{\sigma}{\sqrt{n}} \quad (\sigma \text{ známé})$$

$$\sigma^2 \in \left(\frac{(n-1)s^2}{\chi^2_{1-\alpha/2}(n-1)}, \frac{(n-1)s^2}{\chi^2_{\alpha/2}(n-1)} \right)$$

Testy (K1–K5)

Oboustranný test: kvantil $1-\alpha/2$; jednostranný: $1-\alpha$.

$$T = \frac{\bar{x} - \mu_0}{s}\sqrt{n} \sim t(n-1)$$

$$U = \frac{\bar{x} - \mu_0}{\sigma}\sqrt{n} \sim \mathrm{No}(0,1)$$

$$d_i = x_i - y_i: \quad T = \frac{\bar{d}}{s_d}\sqrt{n} \sim t(n-1)$$

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$$

$$T = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t(n_1+n_2-2)$$

$$U = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}} \approx \mathrm{No}(0,1)$$

$$F = \frac{s_1^2}{s_2^2} \sim F(n_1-1,\, n_2-1), \qquad s_1^2 \geq s_2^2$$

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} \sim \chi^2(n-1)$$

Zamítáme H_0: $|T| \geq t_{1-\alpha/2}$; $T \leq -t_{1-\alpha}$; $T \geq t_{1-\alpha}$; $F \geq F_{1-\alpha/2}$.

Testy dobré shody

$$\chi^2 = \sum_{j=1}^{k} \frac{(n_j - E_j)^2}{E_j}, \qquad E_j = n\,p_j \geq 5, \qquad \chi^2 \geq \chi^2_{1-\alpha}(k-1-r)$$

r = počet odhadnutých parametrů (Po: $\lambda = \bar{x}$, r = 1; No: $\mu = \bar{x}$, σ, r = 2).

$$\text{kontingenční tabulka: } E_{ij} = \frac{r_i\,s_j}{n}, \qquad \nu = (r-1)(s-1)$$

$$\text{K–S: } D = \max_j \left|F_n(x_j) - F_0(x_j)\right|, \qquad F_n(x_j) = \frac{n_1 + \cdots + n_j}{n}$$

$$D \geq D_n(\alpha); \qquad D_n(0{,}05) \approx \frac{1{,}36}{\sqrt{n}}, \quad D_n(0{,}01) \approx \frac{1{,}63}{\sqrt{n}} \quad (n > 40)$$

Regrese a korelace

$$\hat{y} = a + bx, \qquad b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}, \qquad a = \bar{y} - b\,\bar{x}$$

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{\left(n\sum x^2 - (\sum x)^2\right)\left(n\sum y^2 - (\sum y)^2\right)}}$$

Kvantily u_p normovaného normálního rozdělení

p	0,8	0,85	0,875	0,9	0,95	0,975	0,98	0,99	0,995
u_p	0,842	1,036	1,150	1,282	1,645	1,960	2,054	2,326	2,576

Kvantily t a χ^2 (ν = stupně volnosti)

ν	t 0,95	t 0,975	t 0,995	χ^2 0,025	χ^2 0,05	χ^2 0,95	χ^2 0,975
1	6,314	12,706	63,657	0,0010	0,0039	3,841	5,024
2	2,920	4,303	9,925	0,051	0,103	5,991	7,378
3	2,353	3,182	5,841	0,216	0,352	7,815	9,348
4	2,132	2,776	4,604	0,484	0,711	9,488	11,14
5	2,015	2,571	4,032	0,831	1,145	11,07	12,83
6	1,943	2,447	3,707	1,237	1,635	12,59	14,45
7	1,895	2,365	3,499	1,690	2,167	14,07	16,01
8	1,860	2,306	3,355	2,180	2,733	15,51	17,53
9	1,833	2,262	3,250	2,700	3,325	16,92	19,02
10	1,812	2,228	3,169	3,247	3,940	18,31	20,48
11	1,796	2,201	3,106	3,816	4,575	19,68	21,92
12	1,782	2,179	3,055	4,404	5,226	21,03	23,34
13	1,771	2,160	3,012	5,009	5,892	22,36	24,74
14	1,761	2,145	2,977	5,629	6,571	23,68	26,12
15	1,753	2,131	2,947	6,262	7,261	25,00	27,49
16	1,746	2,120	2,921	6,908	7,962	26,30	28,85
17	1,740	2,110	2,898	7,564	8,672	27,59	30,19
18	1,734	2,101	2,878	8,231	9,390	28,87	31,53
19	1,729	2,093	2,861	8,907	10,12	30,14	32,85
20	1,725	2,086	2,845	9,591	10,85	31,41	34,17
21	1,721	2,080	2,831	10,28	11,59	32,67	35,48
22	1,717	2,074	2,819	10,98	12,34	33,92	36,78
23	1,714	2,069	2,807	11,69	13,09	35,17	38,08
24	1,711	2,064	2,797	12,40	13,85	36,42	39,36
25	1,708	2,060	2,787	13,12	14,61	37,65	40,65
26	1,706	2,056	2,779	13,84	15,38	38,89	41,92
27	1,703	2,052	2,771	14,57	16,15	40,11	43,19
28	1,701	2,048	2,763	15,31	16,93	41,34	44,46
29	1,699	2,045	2,756	16,05	17,71	42,56	45,72
30	1,697	2,042	2,750	16,79	18,49	43,77	46,98
40	1,684	2,021	2,704	24,43	26,51	55,76	59,34
50	1,676	2,009	2,678	32,36	34,76	67,50	71,42
60	1,671	2,000	2,660	40,48	43,19	79,08	83,30
80	1,664	1,990	2,639	57,15	60,39	101,88	106,63
100	1,660	1,984	2,626	74,22	77,93	124,34	129,56
120	1,658	1,980	2,617	91,57	95,70	146,57	152,21
∞	1,645	1,960	2,576				

Kritické hodnoty $D_n(\alpha)$ testu K–S

n	1	2	3	4	5	6	7	8	9	10
0,05	0,975	0,842	0,708	0,624	0,563	0,519	0,483	0,454	0,430	0,409
0,01	0,995	0,929	0,829	0,734	0,669	0,617	0,576	0,542	0,513	0,489
n	12	14	16	18	20	25	30	35	40	> 40
0,05	0,375	0,349	0,327	0,309	0,294	0,264	0,242	0,224	0,210	1,36/√n
0,01	0,449	0,418	0,392	0,371	0,352	0,317	0,290	0,269	0,252	1,63/√n